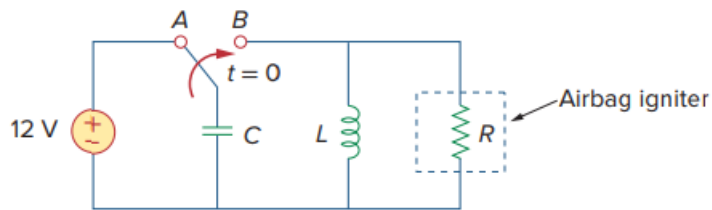


Problem

An automobile airbag igniter is modeled by the circuit in Fig. 8.122. Determine the time it takes the voltage across the igniter to reach its first peak after switching from A to B. Let $R = 3\ \Omega$, $C = 1/30\ \text{F}$, and $L = 60\ \text{mH}$.

Figure 8.122:



Step-by-step solution

Step 1 of 3

The voltage across the igniter is $V_R = V_C$, since the circuit is parallel

RLC n,

$$V_C(0) = 12 \quad \& \quad i_L(0) = 0$$

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 3 \times \frac{1}{30}} = 5$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{60 \times 10^{-3} \times \frac{1}{30}}} = 22.36$$

$\alpha < \omega_0$ produce an under damped response,

Step 2 of 3

$$S_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm j21.749$$

$$V_C(t) = e^{-5t} (A \cos 21.749t + B \sin 21.749t) \rightarrow (1)$$

$$V_C(0) = 12 = A$$

$$\begin{aligned} \frac{dV_C}{dt} &= -5 [A \cos 21.794t + B \sin 21.794t] e^{-5t} \\ &\quad + 21.794 [-A \sin 21.794t + B \cos 21.794t] e^{-5t} \end{aligned}$$

$$\frac{dV_C(0)}{dt} = -5A + 21.79B$$

$$\frac{dV_C(0)}{dt} = \frac{-[V_C(0) + Ri_L(0)]}{RC}$$

$$= \frac{-(12+0)}{\frac{1}{10}} = -120$$

Step 3 of 3

Hence,

$$-120 = -5A + 21.79B$$

$$\text{Leads to } B \frac{(5 \times 12 - 120)}{21.794} = -2.753$$

At the peak value,

$$\frac{dV_c}{dt} = 0,$$

$$0 = A + B \tan 21.794 t_0 + \left(A \frac{21.794}{5} \right) \tan 21.794 t_0$$

$$- 21.794 \frac{B}{5} \left(B + A \frac{21.794}{5} \right) \tan 21.794 t_0 = \left(21.794 \frac{B}{5} \right) A$$

$$\tan 21.794 t_0 = \frac{\left[\left(\frac{21.794 B}{5} \right) + A \right]}{\left[B + \frac{A 21.794}{5} \right]} = \frac{-24}{49.55} = -0.484$$

$$\text{Therefore, } 21.7945 t_0 = |-0.451| t_0 = \frac{|-0.451|}{21.794} = 20.68 \text{ mS}$$